2-Hydroxyethyl Methacrylate

What is it?

2-hydroxyethyl methacrylate (2-HEMA) is a type of methacrylate monomer.

Acrylates are used to make strong glues. They are also used to mold things into plasticized shapes. Major sources of exposure include acrylic and gel nails, dental fillings, and hearing aids.

Acrylates start off as a chemical monomer that is a powder, liquid, gel, or paste. This monomer combines and hardens during a process called polymerization or curing. It is the monomer that can cause you to be allergic. Monomers are less likely to be present after curing. However, there may be leftover monomers if curing is incomplete and this can lead to allergic reactions.

Copolymers should not cause you to be allergic.

2-HYDROXYETHYL METHACRYLATE can also be known as:

- 2-HEMA, HEMA
- Glycol methacrylate
- Glycol monomethacrylate
- Ethylene glycol methacrylate
- 2-Hydroxyethylmethacrylate
- 2-(Methacryloyloxy)ethanol
- Hydroxyethyl methacrylate
- Ethylene glycol monomethacrylate
- 1,2-Ethanediol mono (2-methylpropenoate)
- 2-hydroxyethyl 2-methacrylate
- Methacrylic acid, 2-hydroxyethyl ester

Some Possible Uses of 2-HYDROXYETHYL METHACRYLATE / Where it might be found:

- Artificial sculptured nails, gel nails, shellac gel nail polish
- Tape (non-rubber)
- Adhesive, sealer, and stoppers
- Artificial eye lens implants
- Artificial fingernails
- Bone cement
- Conducting gel for diathermy
- Soft contact lenses
- Dental fillings, crowns, veneers, dentures, prostheses
- Electron microscopy material
- Exfoliating cleanser
- Eye glasses
- Feminine pads
- Floppy disk and hard disk manufacture
- Fragrance base
- Gel electrophoresis
- Glucose monitor
- Hairspray
- Hearing aids
- Incontinence pads
- Industrial adhesive, epoxies
- In-ear headphones
- Insecticide
- Insulin pump
- Leather finish resins
- Light curing gels
- Microporous biomaterials
- Mineral sedimentation
- Nail extender and hardener
- Nail polish (regular, gel, powder dipped)

- Paint plasticizer
- Paper coating
- Plastic foam and coatings
- Plexiglass
- Printing ink, printing plates
- Rubber coating
- Soil improver
- Solvent
- Splints
- TV manufacture
- Textile coatings and sizings
- UV cured glue, ink, paint
- UV adhesive, lacquer
- Water purification
- Wig fixing adhesive
- Wind shield repair
- Wood lacquer
- Wound spray

Other products that may cross react with 2-HYDROXYETHYL METHACRYLATE:

Anything that ends in –acrylate, such as:

- Methyl methacrylate
- Ethyl and polymethyl acrylate
- Ethylene glycol dimethacrylate (EGDMA)
- Diethylene glycol dimethacrylate (DEGDMA)
- Triethylene glycol dimethacrylate (TREGDMA)
- 1,4 - butanediol dimethacrylate (BUDMA)
- Trimethylol propane trimethacrylate
- 1,6 - hexanediol diacrylate (HDDA),
- Triethylene glycol diacrylate (TREGDA)
- Epoxy acrylate

⁎ Copolymers that end in –acrylate should be safe for you to use.

What should I do to avoid 2 - HYDROXYETHYL METHACRYLATE?

Once you are allergic to one acrylate, it is common that your immune system will react to other acrylates even if you have never been exposed to them before (cross-reactions).Thus, avoidance of all monomers is necessary.

Tell your dentist, manicurist, and orthopedic surgeon about this allergy.

This allergen will penetrate rubber gloves in under 60 seconds. Two pairs of neoprene or polyvinyl alcohol gloves may protect for a few minutes. For more protection, try:

- Honeywell Silver Shield / 4H gloves (part# SSG/7) plastic gloves (www.allerderm.com, 800-365-6868 or www.northsafety.com, 800-430-4110)
- Polyvinyl alcohol gloves (Ansell AlphaTec part# 15-554)
- Neoprene gloves (Ansell TouchNTuff #73-300 or #73-500, Ansell GAMMEX surgical gloves #20277255)
- The above gloves could be purchased from (www.grainger.com or www.airgas.com or www.nationwideindustrialsupply.com).

Dentists may use zinc phosphate or zinc oxide cement to avoid acrylates. Dentures can also be boiled in the mold to reduce allergens. Or, substitute prostheses made of thermoplastic nylon, terephthalate, or polyurethane. Use hearing aids made of silicone or polyamide nylon.

Instead of acrylic fingernails, you can try preformed plastic nails, silk wrap or dipped nails with cyanoacrylate glue if you test negative to it, or regular nail polish if there are no acrylates.

In addition, the ACDS Contact Allergen Management Program® (CAMP®) can help manage contact dermatitis. It helps to find products that are safe to use by creating a list of products that do not contain one's allergens. For more information about CAMP®, please ask your doctor or visit www.contactderm.org.

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